



Test Automation Framework (TAF::Perl)

HammerDB TPROC-H Overview/Quick Start

MariaDB Foundation <https://mariadb.org>

Open, vendor-neutral, community-driven tooling for database testing and benchmarking.

TAF-Perl is maintained and published by the MariaDB Foundation to support transparent, reproducible, and fair performance evaluation across the open source database ecosystem. The framework is part of the Foundation's mission to provide open tooling that benefits the entire database community.

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HammerDB TPROC-H Overview

HammerDB TPROC-H is an open workload created and maintained by Steve Shaw. It models the analytical decision-support concepts described in the publicly available TPC-H specification from the Transaction Processing Performance Council.

The TPC-C specification can be found at: <https://www.tpc.org/tpch/>

HammerDB is documented at: <https://www.hammerdb.com/>

Within TAF, the HammerDB TPROC-H test suite provides a deterministic and contributor-proof method for running analytical decision-support workloads across multiple database engines. TAF does not modify HammerDB, does not validate TPC-H compliance, and does not certify results. The suite focuses on lifecycle control, configuration merging, and consistent execution across environments.

The test suite is responsible for initialization, deterministic merging of default properties, user properties, and command-line overrides, execution of the supported test cases, and collection of results for downstream reporting. It enforces stable and reproducible behavior across repeated runs.

The suite does not implement HammerDB itself, does not manage database provisioning, and does not interpret or certify benchmark results. All workload semantics originate from HammerDB, and all analytical concepts originate from the publicly available TPC-H specification.

By integrating HammerDB TPROC-H into TAF, the goal is to provide a consistent and script-driven method for running long-lived analytical workloads across MariaDB, MySQL, PostgreSQL, and other engines. This ensures that performance testing remains transparent, repeatable, and aligned with the architectural principles of TAF.

TAF - Test Suite Structure and Locations

The HammerDB TPROC-H test suite follows the standard TAF layout. All components are located under the TAF installation root.

Test Suite Module: taf-perl/test_suites/hammerdb-tproch.pm

This file defines lifecycle routines, configuration handling, and execution flow.

Default Properties: taf-perl/properties/default/hammerdb_tproch_default.properties

Provides deterministic defaults for all supported databases. Users must not modify this file.

HammerDB Client Source: taf-perl/client_source/hammerdb/HammerDB-5.0/

Contains the HammerDB CLI executable, Tcl scripts, and agent binaries used by the test suite.

Script Directory: taf-perl/scripts/hammerdb/

Contains TAF wrapper scripts and helper logic used during execution.

Configuration Resolution Order

TAF resolves configuration in a strict and deterministic order. This guarantees that every test suite behaves the same way across environments and that users always know which value is in effect. All test suites, including HammerDB TPROC-H, follow the same precedence rules.

1. TAF and HammerDB TPROC-H Default Files

Both TAF and the HammerDB TPROC-H test suite provide default properties files. These files define safe, stable baseline settings and may be overridden, but must never be modified directly. Users override defaults only through user properties files.

2. Auto-Populated Values

When certain settings are left unset, TAF and the HammerDB TPROC-H test suite automatically populate them using installation paths or other environment-derived values. These values apply only when the user has not provided an explicit override.

3. User Properties Files

Users supply a test-case-specific properties file to override defaults. This is the intended method for running TAF. Any key defined in a user properties file replaces both the default value and any auto-populated value.

4. Command-Line Overrides

Command-line options have the highest precedence. Any key specified on the command line replaces the value from the user properties file, the auto-populated value, and the default.

This resolution order ensures predictable, contributor-proof behavior. The final configuration used by the test suite is always the result of applying these four layers in order, with command-line options taking priority over all other sources.

Test Suite Properties

Test suite properties begin with the prefix: **hammerdb_tproch.<option>=**

These properties control workload behavior, client paths, agent settings, and database specific extensions. Defaults come from the test suite's default properties file. Users may override any value through a user properties file or a command-line option.

Examples:

- hammerdb_tproch.scale=1
- hammerdb_tproch.total_querysets=1
- hammerdb_tproch.db_type=maria

TAF Options That Affect Test Suite Execution

The following is a sample of TAF options and properties that impact test suite execution. These options directly influence how the HammerDB TPROC-H test suite runs. Each option has both a command-line options form (O) and a properties form (P).

This is not a complete list. TAF provides many additional options and properties, all of which can be viewed by running:

perl taf.pl --help

Users are encouraged to review the full usage output to understand the complete set of capabilities.

Examples.

Action

O = --action=<action>

P = taf.action=<action>

Controls the overall TAF workflow.

Duration

O = --duration=<seconds|min>

P = taf.duration=<seconds|min>

How long to run each test.

Seconds vs. minutes depends entirely on the test suite implementation.

Default: 30

Iterations

O = --iterations=<number>

P = taf.iterations=<number>

Number of iterations to run for each thread count.

Default: 1

Thread Counts

O = --threads=1

P = taf.threads=1

Defines the list of thread counts to run.

These options combine with the test suite's own properties to determine the final workload behavior. All values are resolved using TAF's deterministic precedence rules: defaults, auto-populated values, user properties, then command-line overrides.

HammerDB-TPROCH Help

The following is example of running test suite help, and the output of the test suite's help.

perl taf.pl --test-suite=hammerdb-tproch --list-test-suites-help

TAF TSM::LoadTestSuite -> : Called

TAF TSM::LoadTestSuite -> : Attempting to load: hammerdb-tproch

TAF TSM::LoadTestSuite -> : Test suite path : /home/jeb/taf-perl/test_suites/hammerdb-tproch.pm

TAF TSM::LoadTestSuite -> : Test suite has completed loading.

TAF TSM::LoadTestSuite -> : Complete

HammerDB TPROC-H Test Suite HELP

Workload Overview:

TPROC-H is an analytical, read-heavy workload derived from TPC-H.

It executes a fixed set of SQL queries against a scale-dependent dataset and reports a geometric mean execution time. Unlike TPROC-C, it has no transaction mix, no think time, and no warehouse-based concurrency model.

Key Concepts:

scale : Size of the dataset. Must match the database build.
total_querysets : Number of full queryset executions to run.
update_sets : Number of update querysets. Must not exceed total_querysets.
refresh_on : Controls refresh behavior: none, stream, or trickle.
trickle_refresh : Enables trickle refresh. Requires refresh_on=trickle.
refresh_verbose : Emits detailed refresh output. May reduce performance.
raise_query_error: true/false. Stops on query errors when enabled.
verbose : true/false. Controls HammerDB CLI verbosity.

Important Invariants:

scale must be a positive integer.

total_querysets must be ≥ 1 .

update_sets must be $\leq$ total_querysets.

refresh_on must be one of: none, stream, trickle.

trickle_refresh=true requires refresh_on=trickle.

refresh_verbose=true may significantly slow execution.

Database-Specific Notes:

pg_degree_of_parallel : Non-negative integer for PostgreSQL parallelism.

mssql_maxdop : Non-negative integer for SQL Server MAXDOP.

cloud_query : aws, azure, or gcp. Meaning varies by database.

Default Tests:

TPROCH

Legal Tests:

TPROCH

Resolved Properties:

hammerdb_tproch.agent = client_source/hammerdb/HammerDB-5.0/agent/agent

hammerdb_tproch.agent_port = 10001

hammerdb_tproch.agent_started_by_ts = 1

hammerdb_tproch.allow_scale_gt_safe = 0

hammerdb_tproch.checksum_after_run = 0

hammerdb_tproch.checksum_after_setup = 0

hammerdb_tproch.client_executable = client_source/hammerdb/HammerDB-5.0/hammerdbcli

hammerdb_tproch.client_script_dir = scripts/hammerdb

hammerdb_tproch.db_type = mariadb

hammerdb_tproch.driver = not defined

hammerdb_tproch.extra_args =

hammerdb_tproch.include_html_metrics = 0
hammerdb_tproch.include_html_result = 0
hammerdb_tproch.include_html_timing = 0
hammerdb_tproch.include_json_metrics = 0
hammerdb_tproch.include_json_result = 0
hammerdb_tproch.include_json_timing = 0
hammerdb_tproch.maria_cloud_query = 0
hammerdb_tproch.maria_tpch_dbase = not defined
hammerdb_tproch.maria_tpch_pass = not defined
hammerdb_tproch.maria_tpch_user = not defined
hammerdb_tproch.mssqls_advanced_stats = 0
hammerdb_tproch.mssqls_colstore = 0
hammerdb_tproch.mssqls_maxdop = 2
hammerdb_tproch.mssqls_odbc_driver = "ODBC Driver 17 for SQL Server"
hammerdb_tproch.mssqls_odbc_dsn = tpch_dsn
hammerdb_tproch.mssqls_partition_orders_and_lineitems = 0
hammerdb_tproch.mssqls_tpch_dbase = not defined
hammerdb_tproch.mssqls_tpch_pass = not defined
hammerdb_tproch.mssqls_tpch_user = not defined
hammerdb_tproch.mssqls_use_bcp = 1
hammerdb_tproch.mysql_cloud_query = 0
hammerdb_tproch.mysql_ob_partition_num = 1
hammerdb_tproch.mysql_ob_tenant_name = tpch
hammerdb_tproch.mysql_obcompat = 0
hammerdb_tproch.mysql_tpch_dbase = not defined
hammerdb_tproch.mysql_tpch_pass = not defined
hammerdb_tproch.mysql_tpch_user = not defined
hammerdb_tproch.pg_cloud_query = 0
hammerdb_tproch.pg_degree_of_parallel = 2
hammerdb_tproch.pg_gpcompat = 0
hammerdb_tproch.pg_gpcompress = 0
hammerdb_tproch.pg_rs_compat = 0
hammerdb_tproch.pg_tpch_dbase = not defined
hammerdb_tproch.pg_tpch_defaultdbase = not defined

hammerdb_tproch.pg_tpch_pass = not defined
hammerdb_tproch.pg_tpch_superuser = not defined
hammerdb_tproch.pg_tpch_superuserpass = not defined
hammerdb_tproch.pg_tpch_user = not defined
hammerdb_tproch.pg_tspace = pg_default
hammerdb_tproch.raise_query_error = 0
hammerdb_tproch.refresh_on = 0
hammerdb_tproch.refresh_verbose = 0
hammerdb_tproch.safe_scale = 100
hammerdb_tproch.scale = 1
hammerdb_tproch.test_client_version = 5.0
hammerdb_tproch.total_querysets = 1
hammerdb_tproch.trickle_refresh = 1000
hammerdb_tproch.update_sets = 1
hammerdb_tproch.verbose = 0

Reference Sites:

<https://www.hammerdb.com>
<https://www.tpc.org/tpch/>
<https://www.mariadb.org>

End of HammerDB TPROC-H HELP

HammerDB-TPROCH Usage Command Example

Basic command-line for executing the provided example MariaDB user's properties-file.

```
perl taf.pl --prop=./properties/mariadb/beta/hammerdb_tproch_beta.properties
```

User Properties File: properties/mariadb/beta/hammerdb_tproch_beta.properties

```
#####
# TAF SETTINGS (GLOBAL)
#
# These settings control TAF behavior for this test run:
# - what action to perform
# - which report plugins to load
# - which test suite to execute
# - how many iterations and threads to run
#
# These settings apply only to this user properties file and override
# TAF defaults and test-suite defaults.
#####

# Action TAF should perform.
# Run "perl taf.pl --list-actions" for the full list.
taf.action=init-start-db-run-tests

# Comma-separated list of report plugins to run after the test.
taf.report_plugin=taf_res_raw_text.pm,text.pm,tproch_queries_chart.pm,chart_and_test_info_
results_tables_html.pm

# Optional comment describing the purpose of this run.
taf.comments="Testing MySQL using HammerDB TPROC-H test suite"

#####
# TAF TEST EXECUTION SETTINGS
#####

# Number of iterations to run for each thread count.
taf.iterations=5

# Name of the test suite to load.
taf.test_suite=hammerdb-tproch

# Classification of this run.
# Valid values: adhoc, investigation, production, rerun, release
taf.test_type=adhoc

# Test case(s) from the suite to execute.
# hammerdb-tproch supports only: TPROCH
taf.tests=TPROCH
```

```

# Comma-separated list of thread counts to test.
taf.threads=1

# Sleep X seconds after test setup.
taf.sleep_after_test_setup=60

# Sleep X seconds after test run.
taf.sleep_after_test_run=20

#####
# DATABASE SETTINGS
#
# These settings control how TAF locates, configures, and connects to the
# database under test. Any value left commented out will fall back to:
# 1. TAF defaults
# 2. auto-populated values (special null handling)
# 3. command-line overrides
#####

# Path to the MariaDB configuration file to feed into mariadb.
# This must be a valid MariaDB config file.
taf.db_config_file=./database_config_files/mariadb/mariadb_simple_2gbp.cnf

# Optional: directory where the database software has been unpacked.
# If omitted, TAF attempts to auto-detect the install directory.
# taf.db_software_install_dir=/home/jeb/taf-perl/database_software_installs/mariadb-
12.3.0-linux-systemd-x86_64/

# Whether to connect using a UNIX socket instead of TCP.
# true = use socket
# false = use TCP (default)
taf.db_clients_use_unix_socket=true

# Optional overrides for database connection parameters.
# Uncomment only if you need to override TAF defaults.
# taf.database=test
# taf.engine=INNODB
# taf.db_user_pass=MariadbPass_@123
# taf.db_user=mariadb_tester
# taf.host=localhost
# taf.db_socket=/home/jeb/taf-perl/tmp/mysql.sock

#####
# HAMMERDB-TPROCH TEST SUITE SETTINGS
#
# These settings apply ONLY to the HammerDB-TPROCH test suite. They control:
# - scale factor for the TPROC-H workload
# - number of querysets (threads) used by HammerDB
# - database type for TCL configuration
# - checksum behavior
# - per-iteration reporting output
#
# These values override the defaults in hammerdb_tproch_default.properties.
#####

```

```

# Scale factor for the TPROC-H workload.
hammerdb_tproch.scale=1

# Number of querysets to run.
hammerdb_tproch.total_querysets=1

# Database type used when generating HammerDB TCL configuration.
hammerdb_tproch.db_type=maria

# Checksum after setup
hammerdb_tproch.checksum_after_setup=true

# Checksum after each run
hammerdb_tproch.checksum_after_run=true

#####
# REPORTING SETTINGS
#
# These toggles control which per-iteration HammerDB reports are generated.
# All reports are written into the "iterations" subdirectory of the results.
#####

# JSON reports
hammerdb_tproch.include_json_timing=true
hammerdb_tproch.include_json_result=true
hammerdb_tproch.include_json_metrics=true

# HTML reports
hammerdb_tproch.include_html_timing=true
hammerdb_tproch.include_html_result=true
hammerdb_tproch.include_html_metrics=true

##### taf Debug #####
taf.tools_debug=true
taf.verbose=true
##### End of Properties #####

```

End of Document

This Quick Start was prepared and published by the MariaDB Foundation <https://mariadb.org>

TAF::Perl is part of the Foundation's commitment to open, vendor-neutral tooling for the database ecosystem. The framework is maintained to support fair benchmarking, reproducible testing, and transparent engineering practices across all database makers.

For contributions, feedback, or questions, please visit the Foundation website or contact the maintainers directly.

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